DEBABRATA SAHU



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PROFILE

Electronics and Communication Engineering undergraduate with research and project experience in digital VLSI, RTL design, FPGA-based systems, and electronic circuit design. Interested in research in VLSI and IC design, hardware-efficient computing, RTL design and verification, FPGA architectures, and semiconductor technologies. Experienced in Verilog, Xilinx Vivado, Cadence Virtuoso, and CST Studio Suite through academic projects, research, and technical training.

EDUCATION

NIST University *2023 – 2027*
B.Tech – Electronics and Communication Engineering(NBA Accredited) — **CGPA: 7.7/10** Berhampur, India

Government Autonomous College *2020 – 2022*
Intermediate – Science — **Percentage: 70%** Rourkela, India

PROJECTS

FPGA-Based Neuromorphic LIF Neuron for Character Recognition *09/2026 – Present*

Currently developing a digital Leaky Integrate-and-Fire (LIF) neuron model for neuromorphic computing and character recognition using Verilog HDL. Exploring spike-based processing and FPGA implementation, with a focus on hardware-efficient neural computation and real-time character recognition.

Configurable Floating-Point FMA Unit using Mixed Precision *02/2026 – 04/2026*

Designed a configurable Floating-Point Fused Multiply-Add (FMA) unit using mixed precision for hardware-efficient floating-point computation. Developed and simulated the RTL implementation using Verilog in Xilinx Vivado, focusing on configurable precision and arithmetic functionality.

Terahertz Metamaterial Absorber for Skin Cancer Detection *05/2026 – 06/2026*

Designed and simulated a terahertz metamaterial absorber-based sensing system using CST Studio Suite for potential skin cancer detection. Analyzed electromagnetic response characteristics of the absorber to study changes associated with abnormal tissue and evaluate sensing behavior.

12-Bit Digital-to-Analog Converter *08/2025 – 09/2025*

Designed and simulated a 12-bit Digital-to-Analog Converter (DAC) using Cadence Virtuoso for Digital-to-Analog signal conversion. Performed schematic-level design and simulation to analyze converter functionality and performance.

Smart Greenhouse Monitor System *05/2025 – 06/2025*

Developed an IoT-based greenhouse monitoring system integrating temperature, humidity, gas, and motion sensors for real-time environmental monitoring. Implemented automated control using servo motors, buzzer, LED, Wi-Fi, and Blynk for remote monitoring and plant-safety alerts.

Smart Mining Helmet *03/2025 – 04/2025*

Developed a sensor-integrated mining helmet for monitoring worker heart rate, temperature, humidity, and hazardous gas conditions. Implemented Wi-Fi-based cloud monitoring using ThingSpeak and Blynk with automated alerts through buzzer, LED, and LCD.

INTERNSHIPS

Research Intern *18 May 2026 – 18 June 2026*
National Institute of Technology Jamshedpur Jamshedpur, Jharkhand

Gained research exposure in terahertz technology, metamaterial-based sensing, and electromagnetic simulation for biomedical applications.

VLSI Project-Based Intern *20 January 2026 – 18 March 2026*
EdXcellence

Gained hands-on exposure to VLSI design, RTL development, digital logic, and hardware-oriented design methodologies using Verilog/SystemVerilog and Xilinx Vivado.

E&T Department Intern *01 June 2025 – 30 June 2025*
Central Coalfields Limited Ranchi, Jharkhand
Gained practical exposure to embedded systems, sensor interfacing, IoT communication, and real-time monitoring through industrial technology applications.

SKILLS

HDL & Programming: Verilog, Python, C, MATLAB, 8086 Assembly
EDA & Simulation Tools: Xilinx Vivado, Cadence Virtuoso, CST Studio Suite, MATLAB, EDA Playground
Digital & Hardware: Digital VLSI Design, RTL Design, RTL Verification, FPGA, Digital Logic Design, Hardware Debugging
Embedded & IoT: Arduino, ESP32, Arduino IDE, Sensor Interfacing, ThingSpeak, Blynk, Microcontrollers, 8086 Microprocessor

TRAINING & CERTIFICATIONS

FPGA-Based Communication System Design (FCSD-2026) *25 May 2026 – 05 June 2026*
Hands-on training in FPGA-based system design, digital communication systems, hardware implementation, and FPGA architecture and development.
Getting Started with Artificial Intelligence *27 February 2026*
Completed introductory training in artificial intelligence concepts and applications through IBM SkillsBuild.
MATLAB & Its Applications in Engineering *12 July 2024 – 01 August 2024*
Gained practical exposure to MATLAB for engineering computation, analysis, simulation, and problem-solving.

WORKSHOPS & TECHNICAL LEADERSHIP

Tech Spark – IoT Workshop *18–19 September 2026*
Served on the organizing team for a two-day hands-on IoT workshop through the Renewable Energy Club, focusing on practical applications of IoT, sensors, and embedded technologies.
SensoTech – Hands-on Electronics Workshop *11–12 April 2025*
Served on the organizing team for a two-day hands-on workshop through the Renewable Energy Club covering sensors, microcontrollers, project building, and practical applications of electronics.
Tinker to Tech – Project Building Workshop *26–27 August 2024*
Participated in a two-day hands-on workshop organized by the Renewable Energy Club in collaboration with the IEEE Student Branch, NIST University, covering sensors, microcontrollers, and basic electronics project development.

RESEARCH PUBLICATIONS

FPGA-Aware Optimization of Approximate Multipliers for Error-Constrained AI Accelerator
D. Sahu, S. K. Verma, A. Anjani, and D. N. Panigrahi, M.Baral in *Proceedings of the 3rd International Conference on Venture Capital and Artificial Intelligence (NGVCAI 2026).*
Communicated for Presentation *6–7 November 2026* • Berhampur, Odisha, India

EXTRACURRICULAR ACTIVITIES

Renewable Energy Club *September 2024 – Present*
Event Head • Social Media Head NIST University, Berhampur
Innovation & Incubation Club *November 2025 – Present*
Core Member NIST University, Berhampur
NIST Basketball Club *October 2023 – Present*
Player NIST University, Berhampur

REFERENCES

Dr. Sandipan Mallik
Professor, Head of Department of Electronics and Communication Engineering
NIST University
✉ sandipan@nist.edu ✆ +91-9658965687

Dr. Ashwini Kumar Nayak
Sr. Assistant Professor, Department of Electrical Engineering
NIST University
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DECLARATION

I hereby declare that the information given above is true and accurate to the best of my knowledge and belief.

Debabrata Sahu

Berhampur, Odisha **Debabrata Sahu**